

Total No. of Questions : 8]

SEAT No. :

P6747

[6181] - 335

[Total No. of Pages : 2

B.E. (Electronics and Telecommunication) (Honors in Robotics)
INDUSTRIAL ROBOTICS AND AUTOMATION
(2019 Pattern) (Semester - VII) (404181 HR)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw neat diagrams wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) All questions carry equal marks.
- 5) Assume suitable data if necessary.

Q1) a) Define pneumatic Actuators, how they are classified. Explain Rotary actuators with neat diagram. **[9]**

b) Explain air treatment stages and pressure regulation method with neat diagram. **[8]**

OR

Q2) a) How pneumatic compressors are classified. Explain positive displacement compressors with neat diagram. **[9]**

b) Explain basic components of pneumatic system with neat diagram. **[8]**

Q3) a) How to perform the analysis of an assembly. Explain with a suitable example. **[9]**

b) Write steps to design of parts for high speed feeding and orienting. State different feeding difficulties. **[9]**

OR

Q4) a) State design features of CNC systems. Explain drive system for CNC machine tools. **[9]**

b) Explain in detail how high speed automation Insertion is possible in robotic assembly. State general rules for product design and automation. **[9]**

P.T.O.

- Q5) a)** What are the stages in mechatronics design. Explain different phases of mechatronics design process. [8]
b) Write a case study on Engine management system. Compare Traditional and mechatronics design. [9]

OR

- Q6) a)** Prepare a detailed case studies on a pick and place robot. Write algorithm for a pick and place robot. [9]
b) Define Mechatronics. Explain basic elements of mechatronics system with neat sketches. [8]
- Q7) a)** Compare electrical, Mechanical and Hydraulic transmission systems. [6]
b) What is back pressure? When it arise and how to avoid it? [6]
c) Draw neat diagram to achieve Forward and Reverse stroke using cylinder and piston mechanism. [6]

OR

- Q8) a)** Give classification of hydraulic pumps. Explain positive displacement pumps and its characteristics. [6]
b) Give comparison between hydraulic and pneumatic system. What are the applications of hydraulic systems. [6]
c) Draw and explain basic components of hydraulic system? [6]

